1 An Open Economy Model

The closed economy model from intermediate macro assumes that the domestic economy is closed, that is, it does not trade goods or financial assets with the rest of the world. We now relax that assumption. We allow the domestic economy to trade goods and services with a foreign economy, which we can think of as the rest of the world, and we allow financial investors to hold domestic and foreign currency and bonds.

Foreign variables are denoted by an asterisk. For example, if Y is domestic output, Y^* denotes foreign output.

1.1 The foreign exchange market

<u>The nominal exchange rate</u>

The **nominal exchange rate** is the price of foreign currency in units of domestic currency, and we denote it by E. With this convention, an increase in E is a **depreciation** of the domestic currency, and a decrease in E is an **appreciation**.

We will use dollars (USD) as the default currency for the domestic economy and euros (EUR) as the default currency for the foreign economy.

Example 1.1. Imagine that the exchange rate between the U.S. dollar and the euro is

$$E_{USD/EUR} = 1.16.$$

Then one euro can be exchanged for 1.16 dollars:

$$\begin{aligned} 1\text{ EUR} \times E_{USD/EUR} &= 1\,\cancel{\text{EUR}} \times 1.16\,\frac{\text{USD}}{\cancel{\text{EUR}}} \\ &= 1.16\text{ USD}. \end{aligned}$$

This means that the price of one euro is 1.16 dollars. Equivalently, the price of one dollar in terms of euros is

$$E_{EUR/USD} = \frac{1}{E_{USD/EUR}} = \frac{1}{1.16}\frac{\text{EUR}}{\text{USD}} = 0.86\,\frac{\text{EUR}}{\text{USD}},$$

so that one dollar can be exchanged for 0.86 euros:

$$1 \text{ USD} \times E_{EUR/USD} = 0.86 \text{ EUR}.$$

If $E_{USD/EUR}$ increases from 1.16 to 1.30, it now takes 1.30 dollars to buy one euro. Since it now takes more dollars than before to buy one euro, the value of the dollar relative to the euro has fallen. This means that the dollar has depreciated with respect to the euro. □

Equilibrium in the foreign exchange market

Investors can buy a domestic bond using dollars and a foreign bond using euros. Investing 1 USD in the domestic bond yields $(1+R)$ USD at maturity, while investing 1 EUR in the foreign bond yields $(1+R^*)$ EUR at maturity. We assume both bonds are one-period bonds that never default. Therefore, R is both the expected and actual return on the domestic bond, measured in dollars, and R^* is both the expected and actual return on the foreign bond, measured in euros.

Let E^e denote the **expected nominal exchange rate**, the best guess of the exchange rate that will prevail next period based only on information available in the current period. The expected return on the foreign bond, measured in dollars, is

$$R^* + \frac{E^e}{E} - 1. \tag{1.1}$$

The term $(E^e/E - 1)$ is the **expected depreciation** of the domestic currency with respect to the foreign currency. To see why equation (1.1) gives the expected return on the foreign bond in dollars, consider investing 1 USD in the foreign bond. The foreign bond must be purchased using euros. At the current exchange rate E, 1 USD can be exchanged for $(1/E)$ EUR. Investing $(1/E)$ EUR in the foreign bond yields a payoff of $(1/E)\times(1+R^*)$ EUR next period. Since the exchange rate expected for next period is E^e, this payoff is expected to be exchanged for an amount of dollars equal to $[(1/E) \times (1 + R^*)] \times E^e$ USD. Since an initial investment of 1 USD in the foreign bond has an expected payoff of $[(1/E) \times (1 + R^*)] \times E^e$ USD, we conclude that the expected return on the foreign bond, measured in dollars, is

$$\frac{[(1/E) \times (1 + R^*)] \times E^e \text{ USD} - 1 \text{ USD}}{1 \text{ USD}} = (1 + R^*) \times \frac{E^e}{E} - 1.$$

Assuming that R^* and $(E^e/E - 1)$ are relatively small, say, no larger than 0.2, this expected return can be accurately approximated by the expression in equation (1.1).[1]

The equilibrium condition in the foreign exchange market is that the domestic and foreign bonds have the same expected returns in dollars:

$$R = R^* + \frac{E^e}{E} - 1. \tag{1.2}$$

The left-hand side is the dollar return on the domestic bond. The right-hand side is the expected dollar return on the foreign bond. Equation (1.2) is called the **uncovered interest parity** condition, or **UIP** for short.

Solving equation (1.2) for E gives

$$E = \frac{E^e}{1 + R - R^*}.$$

For given E^e and R^*, an increase in the domestic interest rate R requires an appreciation of the dollar, that is, a lower E, to raise the expected dollar return on the foreign bond. When the dollar appreciates in the current period, the same initial dollar investment can be exchanged for more euros, resulting in a larger investment in the foreign bond. With R^* unchanged, the larger initial investment implies a larger euro payoff. With E^e unchanged, the larger euro payoff implies a higher expected dollar payoff as well.

[1]

$$\begin{aligned}
(1 + R^*) \times \frac{E^e}{E} - 1 &= (1 + R^*) \times \left(\frac{E^e}{E} - 1 + 1\right) - 1 \\
&= (1 + R^*) \times \left(1 + \left(\frac{E^e}{E} - 1\right)\right) - 1 \\
&= (1 + R^*) + (1 + R^*) \times \left(\frac{E^e}{E} - 1\right) - 1 \\
&= R^* + (1 + R^*) \times \left(\frac{E^e}{E} - 1\right) \\
&= R^* + \left(\frac{E^e}{E} - 1\right) + R^* \times \left(\frac{E^e}{E} - 1\right).
\end{aligned}$$

Ignoring the last term $R^* \times (E^e/E - 1)$ gives the approximation.

1.2 The money market

The money market is exactly the same as in the closed economy, so we simply state the equilibrium condition that **real money supply** equals **real money demand**

$$\frac{M^s}{P} = \mathrm{L}(\underset{(-)}{R}, \underset{(+)}{Y}), \tag{1.3}$$

where M^s is the money supply and $\mathrm{L}(R, Y)$ is the real money demand function, which is decreasing in the nominal interest rate R and increasing in real income Y.

1.3 The market for goods and services

<u>Goods and services</u>

The domestic economy produces many goods and services. We are only interested in aggregate output, so we construct a representative basket of goods and services and treat that basket as the single aggregate good in the domestic economy. Output Y is the number of baskets of domestic goods produced.

The foreign economy has its own representative basket of goods and services with output Y^*.

<u>The price level</u>

The price in dollars of the domestic good is the domestic **price level**, P. The price in euros of the foreign good is the foreign price level, P^*.

1.4 The real exchange rate

The **real exchange rate** is defined by

$$q \equiv \frac{EP^*}{P}. \tag{1.4}$$

The numerator EP^* is the price in dollars of one basket of European goods, and the denominator P is the price in dollars of one basket of American goods. Thus, the real exchange rate q is the price of foreign goods in terms of domestic goods. An increase in q is a **real depreciation**, and a decrease in q is a **real appreciation**.

Example 1.2. Imagine that $E_{USD/EUR} = 1.16$ as in Example 1.1, and that

$$P = 3 \text{ USD per American basket of goods},$$

$$P^* = 8 \text{ EUR per European basket of goods}.$$

The price of one European basket of goods in dollars is

$$\begin{aligned}
EP^* &= E_{\text{USD/EUR}} \times P^* \\
&= 1.16 \frac{\text{USD}}{\text{EUR}} \times \frac{8 \text{ EUR}}{\text{European basket of goods}} \\
&= 1.16 \frac{\text{USD}}{\cancel{\text{EUR}}} \times \frac{8 \cancel{\text{EUR}}}{\text{European basket of goods}} \\
&= 1.16 \times 8 \times \frac{\text{USD}}{\text{European basket of goods}} \\
&= 9.28 \times \frac{\text{USD}}{\text{European basket of goods}}.
\end{aligned}$$

The real exchange rate is

$$\begin{aligned}
q &= \frac{EP^*}{P} \\
&= \frac{9.28 \times \frac{\text{USD}}{\text{European basket of goods}}}{\frac{3 \text{ USD}}{\text{American basket of goods}}} \\
&= \frac{9.28}{3} \times \frac{\text{USD}}{\text{European basket of goods}} \times \frac{1}{\frac{\text{USD}}{\text{American basket of goods}}} \\
&= 3.09 \times \frac{\cancel{\text{USD}}}{\text{European basket of goods}} \times \frac{1}{\frac{\cancel{\text{USD}}}{\text{American basket of goods}}} \\
&= 3.09 \times \frac{\text{American basket of goods}}{\text{European basket of goods}}.
\end{aligned}$$

This means that one European basket of goods can be exchanged for 3.09 American baskets of goods:

$$
\begin{aligned}
1 \text{ European basket of goods} \times q &= 1 \text{ European basket of goods} \times 3.09 \\
&\quad \times \frac{\text{American basket of goods}}{\text{European basket of goods}} \\
&= 1 \text{ } \cancel{\text{European basket of goods}} \times 3.09 \\
&\quad \times \frac{\text{American basket of goods}}{\cancel{\text{European basket of goods}}} \\
&= 3.09 \times \text{American basket of goods}
\end{aligned}
$$

□

1.5 Aggregate demand and equilibrium

Demand

Domestic demand is[2]

$$A \equiv C + I + G. \tag{1.5}$$

Domestic demand A and all of its components C, I, and G, include demand for both domestic and foreign goods.

Exports, denoted by EX, are purchases of domestic goods by foreigners. **Imports**, denoted by IM, are purchases of foreign goods by domestic residents. **Net exports** are

$$NX \equiv EX - IM.$$

The **current account** is net exports plus unilateral transfers. In these notes, we ignore unilateral transfers and therefore set

$$CA \equiv NX,$$

and from now on use CA instead of NX.

[2]Domestic demand is also called absorption, which is why we denote it by A. It is the amount of goods and services "absorbed" by the economy. If $A > Y$, the domestic economy "absorbs" more than it produces and the difference $A - Y > 0$ must come from abroad. If $A < Y$, domestic production is higher than absorption and the output $Y - A > 0$ not "absorbed" domestically must be absorbed by the foreign economy.

The **demand for domestic goods** is domestic demand plus the current account:

$$D \equiv A + CA.$$

The demand for domestic goods D is the demand for a single good. This demand can come from two sources: the domestic economy and the foreign economy. Domestic demand A is a demand for two goods: domestic goods and foreign goods. This demand has a single source: the domestic economy. Adding the current account $CA = EX - IM$ to domestic demand A adds the foreign demand for domestic goods (EX) and removes the domestic demand for foreign goods ($-IM$), so $A + CA$ captures exactly the worldwide (domestic plus foreign) demand for domestic goods D.

Consumption, investment, government spending

Define **disposable income** by

$$Y^d \equiv Y - T. \tag{1.6}$$

We assume **consumption** C is an increasing function of Y^d:

$$C = \mathsf{C}(\underset{(+)}{Y^d}). \tag{1.7}$$

Consumption C includes domestic and foreign goods. We assume that higher disposable income Y^d is associated with an increase in consumption of both domestic and foreign goods.

Investment I and **government spending** G are exogenous.[3]

Exports

We assume that exports are given by a function $\mathrm{EX}(q, Y^*)$ that is increasing in the real exchange rate q and in foreign income Y^*:

$$EX = \mathrm{EX}(\underset{(+)}{q}, \underset{(+)}{Y^*}). \tag{1.8}$$

[3]In intermediate macro, we allow investment to depend negatively on the real borrowing interest rate (real interest rate plus a risk premium) and positively on output. Doing the same in this model would provide more realistic and richer results.

When the real exchange rate q increases, there is a real depreciation. Domestic goods become cheaper relative to foreign goods, increasing foreign demand for domestic goods. When foreign income Y^* increases, foreigners demand more goods of all kinds, including domestic goods.[4]

Imports

To understand imports, it is useful to decompose them into the product of a quantity and a price. Let VOLUME denote the volume (quantity) of imports measured in units of foreign goods. Since one unit of foreign goods can be exchanged for q units of domestic goods (as in Example 1.2), the value of imports expressed in units of domestic goods is

$$IM = q \cdot \text{VOLUME}(\underset{(-)}{q}, \underset{(+)}{Y^d}).$$

We make two natural behavioral assumptions for how the volume of imports VOLUME behaves. First, it depends negatively on q: a real depreciation makes foreign goods more expensive, so fewer are purchased. Second, it depends positively on disposable income Y^d: higher domestic disposable income increases domestic demand for all goods, including foreign goods.

When q increases (a real depreciation), imports IM are affected through two channels. First, a **volume effect**: foreign goods become relatively more expensive, so the quantity VOLUME falls. Second, a **value effect** or **price effect**: the real exchange rate is the price of foreign goods relative to domestic goods, so a real depreciation increases the domestic-good value of each imported unit. Since VOLUME is in units of foreign goods and q has units of domestic goods per foreign goods, multiplying VOLUME by q is simply a change of units. We want to express imports IM in units of domestic goods so that we can combine it with exports EX and the rest of the variables, all of which are in units of domestic goods. This value effect is not a behavioral assumption, it is a mechanical effect arising from the need to write all variables in consistent units.

In response to an increase in the real exchange rate q, the volume effect makes imports go

[4] Using foreign income rather than foreign disposable income is equivalent to keeping foreign taxes unchanged at all times. Since we will not consider changes in foreign taxes, this is without loss of generality.

down, while the value effect makes imports go up.

Example 1.3 (Volume effect dominates). Suppose the volume of imports is

$$\text{VOLUME}(q, Y^d) = \frac{2 + Y^d}{10q} - \frac{1}{10}.$$

The volume is decreasing in q (a real depreciation makes foreign goods more expensive, so fewer are purchased) and increasing in Y^d (higher disposable income raises demand for all goods, including foreign goods). Imports in terms of domestic goods are

$$IM = q \cdot \text{VOLUME}(q, Y^d) = q\left(\frac{2 + Y^d}{10q} - \frac{1}{10}\right) = \frac{2 + Y^d}{10} - \frac{q}{10} = \frac{1}{5} - \frac{1}{10}q + \frac{1}{10}Y^d.$$

Because IM is linear in q and Y^d, we can check if IM is increasing or decreasing in these variables by directly reading off the signs of the coefficients: the coefficient on q is $-1/10 < 0$, so the volume effect dominates. Also, the coefficient on Y^d is $1/10 > 0$, so imports increase with disposable income, as expected.

Picking $Y = 10$, $T = 2$ (so $Y^d = 8$), and $q = 3.09$ from Example 1.2:

$$\text{VOLUME}(3.09, 8) = \frac{2 + 8}{10 \times 3.09} - \frac{1}{10} = \frac{10}{30.9} - 0.1 = 0.224 \text{ baskets of foreign goods,}$$
$$IM = 3.09 \times 0.224 = 0.691 \text{ baskets of domestic goods.}$$

We can check using the formula for IM we derived above:

$$IM = \frac{1}{5} - \frac{1}{10}(3.09) + \frac{1}{10}(8) = 0.2 - 0.309 + 0.8 = 0.691.$$

If q rises from 3.09 to 3.19:

$$\text{VOLUME}(3.19, 8) = \frac{10}{31.9} - 0.1 = 0.214,$$
$$IM = \frac{1}{5} - \frac{1}{10}(3.19) + \frac{1}{10}(8) = 0.681.$$

The volume of imports falls from 0.224 to 0.214 (volume effect), yet the price of each imported unit rises because q increases (value effect). Because the volume effect dominates, imports IM fall (from 0.691 to 0.681). $\square$

Example 1.4 (Value effect dominates). Now suppose the volume of imports is:

$$\text{VOLUME}(q, Y^d) = \frac{2 + Y^d}{10q} + \frac{1}{10}.$$

The volume is still decreasing in q and increasing in Y^d, but the constant term is positive rather than negative, as in Example 1.3. Imports are

$$IM = q \cdot \text{VOLUME}(q, Y^d) = \frac{2 + Y^d}{10} + \frac{q}{10} = \frac{1}{5} + \frac{1}{10}q + \frac{1}{10}Y^d.$$

Now the coefficient on q is $+1/10 > 0$: the value effect dominates. The volume still falls when q rises, but the increase in the domestic-good price of each imported unit more than offsets the decrease in quantity.

Using the same $Y = 10$, $T = 2$, $q = 3.09$ as in Example 1.3:

$$\begin{aligned} \text{VOLUME}(3.09, 8) &= \frac{10}{30.9} + 0.1 = 0.424 \text{ baskets of foreign goods,} \\ IM &= \frac{1}{5} + \frac{1}{10}(3.09) + \frac{1}{10}(8) \\ &= 0.2 + 0.309 + 0.8 = 1.309 \text{ baskets of domestic goods.} \end{aligned}$$

If q rises to 3.19, then

$$\begin{aligned} \text{VOLUME}(3.19, 8) &= \frac{10}{31.9} + 0.1 = 0.414 \qquad \text{(volume falls),} \\ IM &= 0.2 + \frac{1}{10}(3.19) + 0.8 = 1.319 \qquad \text{(imports rise).} \end{aligned}$$

The value effect dominates. □

<u>Current account</u>

From now on, we assume that the volume effect dominates, so that IM is decreasing in q:

$$IM = \text{IM}(\underset{(-)}{q}, \underset{(+)}{Y^d}). \tag{1.9}$$

Using equations (1.8) and (1.9), the current account is

$$CA = \text{EX}(\underset{(+)}{q}, \underset{(+)}{Y^*}) - \text{IM}(\underset{(-)}{q}, \underset{(+)}{Y^d}),$$

so we can write it as a function CA that is increasing in the real exchange rate q, decreasing in domestic disposable income Y^d, and increasing in foreign income Y^*:

$$CA = \text{CA}(\underset{(+)}{q}, \underset{(-)}{Y^d}, \underset{(+)}{Y^*}). \tag{1.10}$$

Example 1.5. From Example 1.3, the import function (with volume effect dominating) is

$$\text{IM}(q, Y^d) = \frac{1}{5} - \frac{1}{10}q + \frac{1}{10}Y^d.$$

Suppose the export function is

$$\text{EX}(q, Y^*) = \frac{1}{10} + \frac{1}{10}q + \frac{1}{10}Y^*.$$

Exports are increasing in q (a real depreciation makes domestic goods cheaper for foreigners) and increasing in Y^* (higher foreign income raises foreign demand for all goods, including domestic goods).

The current account is

$$\begin{aligned}
CA &= EX - IM \\
&= \left(\frac{1}{10}q + \frac{1}{10}Y^* + \frac{1}{10}\right) - \left(-\frac{1}{10}q + \frac{1}{10}Y^d + \frac{1}{5}\right) \\
&= \frac{1}{10}q + \frac{1}{10}Y^* + \frac{1}{10} + \frac{1}{10}q - \frac{1}{10}Y^d - \frac{1}{5} \\
&= -\frac{1}{10} + \underset{(+)}{\frac{1}{5}}q - \underset{(-)}{\frac{1}{10}}Y^d + \underset{(+)}{\frac{1}{10}}Y^*.
\end{aligned}$$

The signs of the coefficients confirm the properties of the abstract function CA: the coefficient on q is $+1/5 > 0$ (increasing in q), the coefficient on Y^d is $-1/10 < 0$ (decreasing in Y^d), and the coefficient on Y^* is $+1/10 > 0$ (increasing in Y^*).

It is important to distinguish CA (the variable) from CA (the function). The variable CA is a number: it is the value of the current account for specific values of q, Y^d, and Y^*. The function CA is the *rule* that maps any triple (q, Y^d, Y^*) to the corresponding current account value. In this example, the function is

$$\text{CA}(q, Y^d, Y^*) = -\frac{1}{10} + \frac{1}{5}q - \frac{1}{10}Y^d + \frac{1}{10}Y^*,$$

and the variable satisfies $CA = \mathsf{CA}(q, Y^d, Y^*)$. Using $q = 3.09$, $Y^d = 8$, $Y^* = 10$, we have:

$$CA = \mathsf{CA}(3.09, 8, 10) = -\frac{1}{10} + \frac{1}{5}(3.09) - \frac{1}{10}(8) + \frac{1}{10}(10) = -0.1 + 0.618 - 0.8 + 1.0 = 0.718.$$

□

Equilibrium

Using equations (1.5), (1.6), (1.7), and (1.10), demand for domestic goods is

$$D = \mathsf{C}(\underset{(+)}{Y^d}) + I + G + \mathsf{CA}(\underset{(+)}{q}, \underset{(-)}{Y^d}, \underset{(+)}{Y^*}).$$

Since C is increasing in Y^d and CA is decreasing in Y^d, it may at first seem that whether D is increasing or decreasing in Y^d depends on the details of how C and CA depend on Y^d. This is not the case. The demand for domestic goods D is always increasing in Y. Consider an increase in Y of one unit. Consumption rises by the marginal propensity to consume (MPC), which is less than one. The increase is split between domestic and foreign goods. The increase in income also leads to an increase in imports. Indeed, the increase in consumption of foreign goods must be exactly equal to the increase in imports, since the additional consumption of foreign goods can only be obtained by buying them from the foreign country, that is, through imports. Therefore, part of the increase in C and all of the decrease in CA cancel each other out. Overall, D still increases, since consumption of domestic goods increases.

Another way to understand why D must increase when Y increases is to note that D is, by definition, the demand for domestic goods only. Changes in the demand for foreign goods are not part of D, so if imports increase, that increase is always "purged" from D by the corresponding subtraction of imports.[5] Since D is unambiguously increasing in Y^d, and

[5]Write consumption, investment, and government spending as sums of domestic and foreign goods:

$$\begin{aligned} C &= C_{\text{domestic}} + C_{\text{foreign}}, \\ I &= I_{\text{domestic}} + I_{\text{foreign}}, \\ G &= G_{\text{domestic}} + G_{\text{foreign}}, \end{aligned}$$

where all the variables are measured in units of domestic goods. We treat I_{domestic}, I_{foreign}, G_{domestic}, and G_{foreign} as exogenous and fixed.

since $Y^d = Y - T$, we can write it as

$$D = \mathrm{D}(\underset{(+)}{q}, \underset{(+)}{Y}, \underset{(-)}{T}, \underset{(+)}{I}, \underset{(+)}{G}, \underset{(+)}{Y^*}). \tag{1.11}$$

where D is a function that is decreasing in taxes T and increasing in all other arguments.

The demand curve in the open economy in equation (1.11) is flatter than in a closed economy. In a closed economy, when income Y rises by one unit, demand rises by an amount equal to the MPC. In an open economy, demand for domestic goods rises by less than the MPC, because part of the increase in consumption goes to imported goods rather than domestic goods. The fraction of each extra unit of income that "leaks" to imports reduces the slope of the demand curve relative to the closed economy. The marginal propensity to import captures this leakage: the higher the marginal propensity to import, the flatter the demand curve and the smaller the open-economy multiplier. Equivalently, the higher the share of foreign goods in consumption, the flatter the demand curve.

Equilibrium in the market for domestic goods requires that the supply of domestic goods Y equals the demand for domestic goods D:

$$Y = D. \tag{1.12}$$

Together with equation (1.11), equation (1.12) gives

$$Y = \mathrm{D}(\underset{(+)}{q}, \underset{(+)}{Y}, \underset{(-)}{T}, \underset{(+)}{I}, \underset{(+)}{G}, \underset{(+)}{Y^*}). \tag{1.13}$$

Solving equation (1.13) for Y gives the equilibrium level of Y as a function of q, T, I, G, and Y^*.

Imports in units of domestic goods are

$$IM = C_{\text{foreign}} + G_{\text{foreign}} + I_{\text{foreign}}.$$

Demand for domestic goods is

$$\begin{aligned} D &= A + CA \\ &= (C + I + G) + (EX - IM) \\ &= (C_{\text{domestic}} + C_{\text{foreign}}) + (I_{\text{domestic}} + I_{\text{foreign}}) + (G_{\text{domestic}} + G_{\text{foreign}}) + EX - (C_{\text{foreign}} + G_{\text{foreign}} + I_{\text{foreign}}) \\ &= C_{\text{domestic}} + I_{\text{domestic}} + G_{\text{domestic}} + EX, \end{aligned}$$

which does not depend on C_{foreign}, as the C_{foreign} component in C cancels out with the C_{foreign} component in IM. When income Y goes up, both C_{domestic} and C_{foreign} increase. If C_{domestic} goes up by $\Delta C_{\text{domestic}}$, then demand for domestic goods D also goes up by $\Delta C_{\text{domestic}}$, since C_{foreign} does not enter D.

Example 1.6. Continue from Example 1.5. Suppose the consumption function is

$$\mathrm{C}(Y^d) = \frac{1}{2}Y^d.$$

The marginal propensity to consume is MPC $= 1/2$. Since $Y^d = Y - T$, domestic demand is

$$A = C + I + G = \frac{1}{2}(Y - T) + I + G.$$

Using the current account from Example 1.5, demand for domestic goods is

$$\begin{aligned} D &= A + CA \\ &= \frac{1}{2}(Y-T) + I + G + \frac{1}{5}q - \frac{1}{10}(Y-T) + \frac{1}{10}Y^* - \frac{1}{10} \\ &= -\frac{1}{10} + \frac{1}{5}q + \frac{2}{5}Y - \frac{2}{5}T + I + G + \frac{1}{10}Y^*. \end{aligned}$$

The function D is therefore

$$\mathrm{D}(\underset{(+)}{q}, \underset{(+)}{Y}, \underset{(-)}{T}, \underset{(+)}{I}, \underset{(+)}{G}, \underset{(+)}{Y^*}) = -\frac{1}{10} + \underset{(+)}{\frac{1}{5}}q + \underset{(+)}{\frac{2}{5}}Y - \underset{(-)}{\frac{2}{5}}T + \underset{(+)}{I} + \underset{(+)}{G} + \underset{(+)}{\frac{1}{10}}Y^*.$$

As with CA versus CA, the variable D is a number (the value of demand for domestic goods), while D is the function that maps (q, Y, T, I, G, Y^*) to that number.

The signs confirm the properties stated above: the coefficient on q is $+1/5 > 0$, on Y is $+2/5 > 0$, on T is $-2/5 < 0$, and on I, G, and Y^* are all positive.

The slope of the demand curve (the coefficient on Y) is $2/5$. In a closed economy with the same MPC $= 1/2$, demand would be $D = \frac{1}{2}Y + \ldots$, so the slope would be $1/2$. Because $2/5 < 1/2$, the open-economy demand curve is flatter. The difference $1/2 - 2/5 = 1/10$ is exactly the marginal propensity to import. □

Example 1.7. Continuing from Example 1.6, set $Y = D$:

$$Y = -\frac{1}{10} + \frac{1}{5}q + \frac{2}{5}Y - \frac{2}{5}T + I + G + \frac{1}{10}Y^*.$$

Collecting terms in Y:

$$Y - \frac{2}{5}Y = \frac{3}{5}Y = -\frac{1}{10} + \frac{1}{5}q - \frac{2}{5}T + I + G + \frac{1}{10}Y^*.$$

Solving for Y:

$$Y = -\frac{1}{6} + \frac{1}{3}q - \frac{2}{3}T + \frac{5}{3}I + \frac{5}{3}G + \frac{1}{6}Y^*.$$

This is the equilibrium level of Y as a function of q, T, I, G, and Y^*. The open-economy multiplier on I and G is $5/3 = 1/(1-2/5) = 1/(1-\text{MPC}+\text{marginal propensity to import})$, compared with the closed-economy multiplier $1/(1-\text{MPC}) = 1/(1-1/2) = 2 > 5/3$. □

1.6 The short-run AA-DD model

The AA schedule

Combining the uncovered interest parity condition (1.2) with the money-market equilibrium condition in (1.3) gives

$$\frac{M^s}{P} = \mathrm{L}\left(R^* + \frac{E^e}{E} - 1, Y\right). \tag{1.14}$$

Equation (1.14) is the **AA schedule**. The **AA curve** is the locus of (Y, E) pairs that satisfy (1.14). Solving for E gives the equation for the AA curve explicitly:

$$E = \mathrm{AA}(\underset{(-)}{Y}; \underset{(-)}{P}, \underset{(+)}{E^e}, \underset{(+)}{M^s}, \underset{(+)}{R^*}). \tag{1.15}$$

where we interpret the function AA as a function of Y whose shape depends on the arguments P, E^e, M^s, and R^*. Changes in P, E^e, M^s, and R^* shift the AA curve. A fall in P, or a rise in E^e, M^s, or R^*, shifts the AA schedule upward. Keeping P, E^e, M^s, and R^* fixed, joint changes in E and Y that continue to satisfy equation (1.15) are movements along the AA curve.

The AA schedule is downward sloping. Fix P, E^e, M^s, and R^*, then find a pair (Y, E) that satisfies equation (1.15). Starting from that pair, consider an increase in Y. When real income Y increases, real money demand goes up. Real money supply does not change, since M^s and P are being kept fixed. Therefore, the interest rate R must increase to make bonds more attractive relative to money, and exactly offset the increase in the real money demand caused by the higher Y. In the foreign exchange market, equilibrium requires that the dollar return on the foreign bond increases to match the increase in the domestic interest rate R. Since E^e and R^* are fixed, the increase in the dollar return on the foreign bond must be

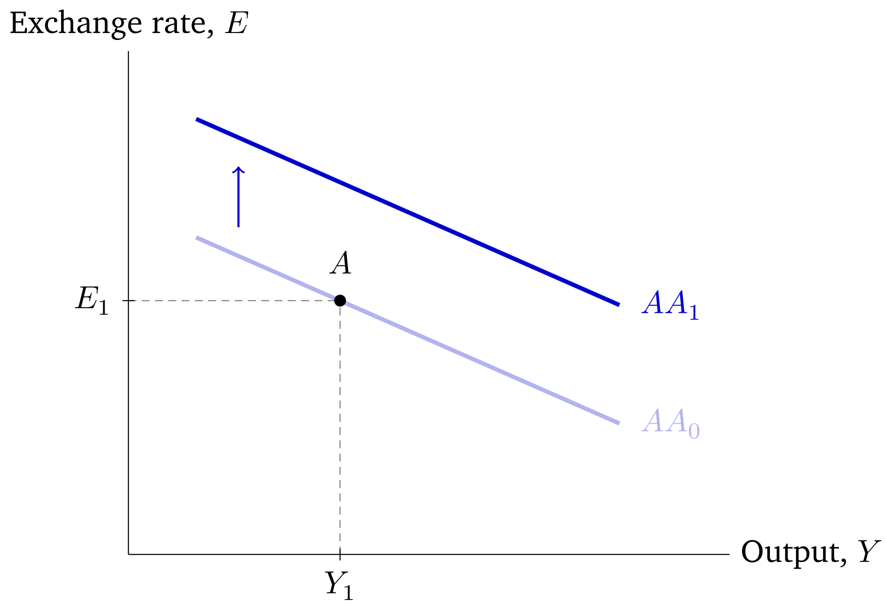

Figure 1.1: The AA schedule and an upward shift. The initial AA curve (AA_0) passes through point A. An increase in M^s, E^e, or R^*, or a decrease in P, shifts the curve upward to AA_1. For any given value of Y, AA_1 gives a higher E than AA_0.

achieved by a lower E, that is, by an appreciation.

We can "trace" the entire AA curve by considering different values for Y, finding the corresponding E using equation (1.15), and plotting the resulting (Y, E) point in the diagram.

Figure 1.1 plots the AA schedule and illustrates an upward shift caused by an increase in M^s, E^e, or R^*, or a decrease in P.

We construct the AA curve using the equilibrium conditions for the money and foreign exchange markets. In those two markets, Y is exogenous and E is endogenous. We therefore interpret the AA curve as providing the equilibrium value of E for a given value of Y. This is why shifts in the AA curve are shifts up and down rather than to the left or to the right (upward and downward shifts leave Y unchanged).

For given values of P, E^e, M^s, and R^*, the following statements are equivalent:

- the point (Y_1, E_1) is on the AA curve;
- in equation (1.15), plugging in $Y = Y_1$ gives $E = E_1$;

- the pair (Y_1, E_1) satisfies equation (1.14);
- if $Y = Y_1$ and $E = E_1$, the money and foreign exchange markets are in equilibrium.

In the last bullet point, we have implicitly assumed that the interest rate R adjusts to a value compatible with equilibrium. Given a pair (Y_1, E_1) on the AA curve, we can find the R that is compatible with equilibrium by plugging $E = E_1$ into the UIP equation (1.2), or $Y = Y_1$ into the money market equilibrium condition in equation (1.3) and solving for R (both give the same R).

This means that even though the interest rate R does not itself show up in the equation of the AA curve, each point in the AA curve is associated with a different equilibrium value of R. Along the AA curve, R is increasing in Y.

Example 1.8. Suppose the real money demand function $\mathrm{L}(\cdot, \cdot)$ is given by

$$\mathrm{L}(R, Y) = \frac{Y}{1+R}.$$

Plugging this money-demand function into (1.14) gives the AA schedule

$$\frac{M^s}{P} = \frac{Y}{R^* + E^e/E}.$$

Solving this equation for E gives the AA curve explicitly:

$$E = \frac{\left(\frac{M^s}{P}\right) E^e}{Y - \left(\frac{M^s}{P}\right) R^*},$$

The numerator is positive and the denominator is increasing in Y, so the AA curve is downward sloping. For any given Y, lower P, higher E^e, higher M^s, or higher R^* raise E, so they cause an upward shift in the AA curve.

Comparing with equation (1.15), the function AA is

$$\mathrm{AA}(Y; P, E^e, M^s, R^*) = \frac{\left(\frac{M^s}{P}\right) E^e}{Y - \left(\frac{M^s}{P}\right) R^*}.$$

For example, with $M^s = 1$, $P = 27/25$, $E^e = 1$, and $R^* = 2/25$:

$$\text{AA}\left(Y; \frac{27}{25}, 1, 1, \frac{2}{25}\right) = \frac{\frac{25}{27}}{Y - \frac{25}{27} \cdot \frac{2}{25}} = \frac{25/27}{Y - 2/27} = \frac{25}{27Y - 2}.$$

□

<u>The DD schedule</u>

Solving the goods-market equilibrium condition in (1.13) for the real exchange rate gives

$$q = \text{Q}(\underset{(+)}{Y}, \underset{(+)}{T}, \underset{(-)}{I}, \underset{(-)}{G}, \underset{(-)}{Y^*})$$

for some function Q that is increasing in Y and T, and decreasing in I, G, and Y^*. Using the definition of the real exchange rate in equation (1.4), and solving for the nominal exchange rate E gives

$$E = \frac{P}{P^*}\text{Q}(Y, T, I, G, Y^*). \tag{1.16}$$

Equation (1.16) is the **DD schedule**. The **DD curve** is the locus of (Y, E) pairs that satisfy (1.16). We can write the equation for the DD curve as

$$E = \text{DD}(\underset{(+)}{Y}; \underset{(+)}{P}, \underset{(+)}{T}, \underset{(-)}{I}, \underset{(-)}{G}, \underset{(-)}{Y^*}, \underset{(-)}{P^*}), \tag{1.17}$$

where we interpret the function DD as a function of Y whose shape depends on the arguments P, T, I, G, Y^*, and P^*. Changes in P, T, I, G, Y^*, and P^* shift the DD curve. A fall in P or T, or a rise in I, G, Y^*, or P^*, shifts the DD schedule to the right. Changes in E and Y, holding P, T, I, G, Y^*, and P^* fixed, are movements along the DD curve.

The DD schedule is upward sloping. For given values of P and P^*, a nominal depreciation (an increase in E) leads to a real depreciation (an increase in q). Exports EX increase, imports IM decrease, and the current account CA improves (increases). Therefore, demand for domestic goods D increases. In equilibrium, output Y must increase so that supply matches the increased demand for domestic goods. Higher output also means higher income, which increases domestic demand, and thus also demand for domestic goods. This further increase in demand for domestic goods increases output in equilibrium, which increases income, and so on.

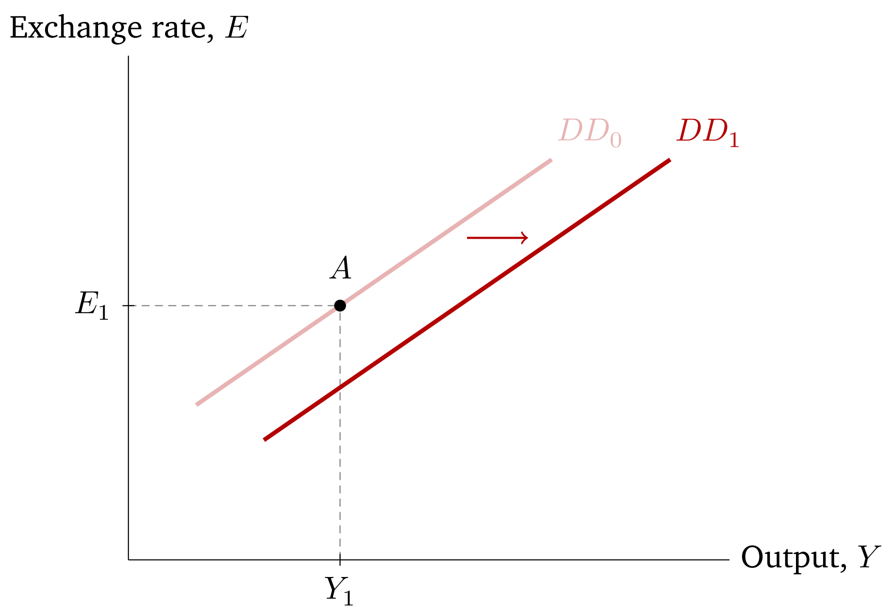

Figure 1.2: The DD schedule and a rightward shift. The initial DD curve (DD_0) passes through point A. An increase in I, G, Y^*, or P^*, or a decrease in P or T, shifts the curve to the right to DD_1. For any given value of E, DD_1 gives a higher Y than DD_0.

Figure 1.2 plots the DD schedule and illustrates a rightward shift caused by an increase in I, G, Y^*, or P^*, or a decrease in P or T.

We construct the DD curve using the equilibrium condition for the goods market. In the goods market, E is exogenous and Y is endogenous. We therefore interpret the DD curve as providing the equilibrium value of Y for a given value of E. This is why shifts in the DD curve are shifts to the left and to the right rather than up or down (leftward and rightward shifts leave E unchanged).

For given values of P, T, I, G, Y^*, and P^*, the following statements are equivalent:

- the point (Y_1, E_1) is on the DD curve;
- in equation (1.17), plugging in $Y = Y_1$ gives $E = E_1$;
- the pair (Y_1, E_1) satisfies equation (1.16);
- if $Y = Y_1$ and $E = E_1$, the goods market is in equilibrium.

Example 1.9. Using the functional forms from Examples 1.6 and 1.7:

$$\begin{aligned}
\mathrm{C}(Y^d) &= \frac{1}{2}Y^d, \\
\mathrm{EX}(q, Y^*) &= \frac{1}{10} + \frac{1}{10}q + \frac{1}{10}Y^*, \\
\mathrm{IM}(q, Y^d) &= \frac{1}{5} - \frac{1}{10}q + \frac{1}{10}Y^d.
\end{aligned}$$

Demand for domestic goods is (from Example 1.6)

$$D = -\frac{1}{10} + \frac{1}{5}q + \frac{2}{5}Y - \frac{2}{5}T + I + G + \frac{1}{10}Y^*.$$

Setting $Y = D$ and solving for q gives

$$\begin{aligned}
\frac{3}{5}Y &= \frac{1}{5}q - \frac{2}{5}T + I + G + \frac{1}{10}Y^* - \frac{1}{10}, \\
q &= \frac{1}{2} + 3Y + 2T - 5I - 5G - \frac{1}{2}Y^*.
\end{aligned}$$

This is the function Q:

$$\mathrm{Q}(Y, T, I, G, Y^*) = \frac{1}{2} + 3Y + 2T - 5I - 5G - \frac{1}{2}Y^*.$$

As required, Q is increasing in Y (coefficient $+3$) and T (coefficient $+2$), and decreasing in I, G (coefficient -5), and Y^* (coefficient $-1/2$).

The DD curve is $E = (P/P^*)\,\mathrm{Q}(Y, T, I, G, Y^*)$:

$$\begin{aligned}
E &= \frac{P}{P^*}\left(\frac{1}{2} + 3Y + 2T - 5I - 5G - \frac{1}{2}Y^*\right) \\
&= \left(\frac{1}{2} + 2T - 5I - 5G - \frac{1}{2}Y^*\right)\frac{P}{P^*} + \frac{3P}{P^*}Y.
\end{aligned}$$

The intercept is $(1/2+2T-5I-5G-Y^*/2)(P/P^*)$ and the slope is $3P/P^* > 0$, confirming the DD schedule is upward sloping. The function DD can be read off directly from the last equation:

$$\mathrm{DD}(Y; P, T, I, G, Y^*, P^*) = \left(\frac{1}{2} + 2T - 5I - 5G - \frac{1}{2}Y^*\right)\frac{P}{P^*} + \frac{3P}{P^*}Y.$$

The coefficient $1/2$ on Y^d in $\mathrm{C}(Y^d)$ is the MPC. The coefficient $1/10$ on Y^d in $\mathrm{IM}(q, Y^d)$ is the marginal propensity to import. The open-economy multiplier is $5/3 = 1/(1 - MPC +$ marginal propensity to import$) = 1/(1 - 1/2 + 1/10)$. The closed-economy multiplier (setting the marginal propensity to import to zero) would be $1/(1-1/2) = 2 > 5/3$, confirming that import leakage reduces the multiplier.

To see how the DD shifts, note that:

- A fall in T lowers the intercept of the DD curve, shifting it to the right (for a given E, a higher Y is now needed to be on the DD curve). A fall in P also shifts the DD curve to the right.[6]
- A rise in I, G, Y^*, or P^* also shifts the DD schedule to the right: higher I, G, or Y^* lower the intercept.

In addition to changing the intercept, higher P or lower P^* also raise the slope of the DD, making the line steeper. When we analyze how the DD changes when P or P^* change, we usually ignore the change in the slope, as it complicates the analysis while leaving all qualitative behavior the same. □

Short-run equilibrium as the intersection of AA and DD

In the models for the money and foreign exchange markets, Y is exogenous and E is endogenous. In the model for the goods market, the reverse is true: E is exogenous and Y is endogenous.

We now consider a new model that combines all three markets. When all three markets are combined, Y and E are both endogenous and jointly determined by the AA and DD curves. The **short-run equilibrium** is an equilibrium in which the money, foreign exchange, and goods markets are all in equilibrium.

For given values of P, E^e, M^s, R^*, T, I, G, Y^*, and P^*, the following statements are equivalent:

[6] A fall in P changes both the intercept and the slope of the DD curve, so the rightward shift cannot be seen from the intercept alone. To verify the rightward shift, solve the DD curve $E = F\frac{P}{P^*} + \frac{3P}{P^*}Y$, where $F = \frac{1}{2} + 2T - 5I - 5G - \frac{1}{2}Y^*$, for Y:

$$Y = \frac{EP^*}{3P} - \frac{F}{3}.$$

When P falls, the first term $\frac{EP^*}{3P}$ increases while the second term $-\frac{F}{3}$ does not depend on P. Therefore, for a given E, equilibrium Y increases, confirming the rightward shift.

- the point (Y_1, E_1) is at the intersection of the AA and DD curves;
- the pair (Y_1, E_1) satisfies equations (1.15) and (1.17) simultaneously;
- if $Y = Y_1$ and $E = E_1$, the money, foreign exchange, and goods markets are all in equilibrium.

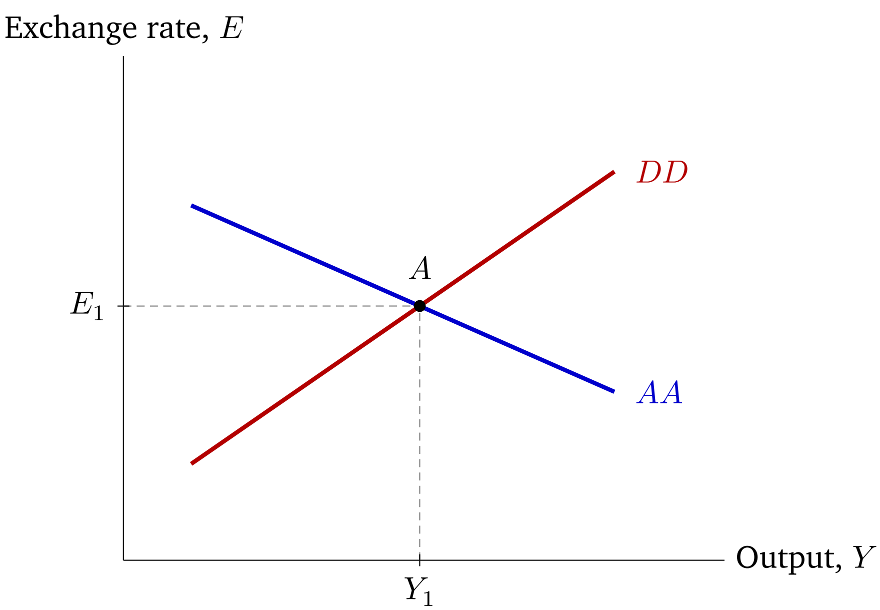

Figure 1.3: Short-run equilibrium in the AA-DD model. The equilibrium is at point A, the intersection of the downward-sloping AA curve and the upward-sloping DD curve. At point A, output is Y_1 and the exchange rate E_1.

Example 1.10. From Example 1.8, the AA curve is

$$E = \frac{\left(\frac{M^s}{P}\right) E^e}{Y - \left(\frac{M^s}{P}\right) R^*}.$$

From Example 1.9, the DD curve is

$$E = \left(\frac{1}{2} + 2T - 5I - 5G - \frac{1}{2}Y^*\right)\frac{P}{P^*} + \frac{3P}{P^*}Y.$$

Using the initial values from Section 1.10 ($T = 0$, $I = G = 1/5$, $Y^* = 1$, $P = P^* = 27/25$,

$M^s = 1$, $E^e = 1$, $R^* = 2/25$), the system becomes

$$\text{AA:} \quad E = \frac{25}{27Y - 2},$$
$$\text{DD:} \quad E = 3Y - 2.$$

To find the short-run equilibrium, equate the E from both expressions:

$$\frac{25}{27Y - 2} = 3Y - 2.$$

Multiplying both sides by $27Y - 2$,

$$25 = (3Y - 2)(27Y - 2) = 81Y^2 - 60Y + 4.$$

Rearranging and simplifying,

$$27Y^2 - 20Y - 7 = 0,$$

which has solution $Y = 1$.[7]

The equilibrium exchange rate E can be found by plugging $Y = 1$ into either the AA or DD (both give the same answer). Using the DD curve, $E = -2 + 3(1) = 1$. The equilibrium is therefore $(Y, E) = (1, 1)$.

The interest rate R can be found from either the money market equilibrium condition or from UIP (both give the same answer). Using the money market with $\mathrm{L}(R, Y) = Y/(1+R)$, $M^s = 1$, and $P = 27/25$:

$$\frac{M^s}{P} = \frac{Y}{1+R} \quad \Longrightarrow \quad 1 + R = \frac{YP}{M^s} = \frac{1 \cdot 27/25}{1} = \frac{27}{25} \quad \Longrightarrow \quad R = \frac{2}{25}.$$

[7] In this course, you will not be asked to solve quadratic equations. However, we show that using the quadratic formula does produce $Y = 1$:

$$Y = \frac{20 + \sqrt{400 + 756}}{54} = \frac{20 + \sqrt{1156}}{54} = \frac{20 + 34}{54} = 1.$$

The second solution of the quadratic is

$$Y = \frac{20 - \sqrt{1156}}{54} = \frac{20 - 34}{54} = -\frac{7}{27} < 0,$$

which is negative and therefore not a valid solution for Y.

Using UIP with $R^* = 2/25$, $E^e = 1$, and $E = 1$:

$$R = R^* + \frac{E^e}{E} - 1 = \frac{2}{25} + \frac{1}{1} - 1 = \frac{2}{25}.$$

□

1.7 Dynamic AA-DD model

We now consider a dynamic version of the AA-DD that allows us to study how the economy evolves over time. To do so, we make the price level P and the expected exchange rate E^e endogenous.

Time dynamics

Time runs continuously. Within this continuous time evolution, we distinguish three particular periods:

- The **initial period** is where the economy begins. We use a subscript 0 for the value of variables in this initial period. For example, the exchange rate in the initial period is E_0.
- The **short run** is the instant of time in which a shock (an unexpected change to an exogenous variable) occurs. We use a subscript SR for variables in the short run, for example, E_{SR}.
- The **long run** is when the economy has finished adjusting to the unexpected changes that occurred in the short run, assuming no further unexpected changes take place. We use a subscript LR for variables in the long run, for example, E_{LR}.

A long run equilibrium is a theoretical construct, a hypothetical equilibrium that occurs when enough time has passed after the initial shock *in the absence of any other shocks*. In the real world, shocks hit the economy all the time, so there is never enough time for all variables to fully adjust to a shock before a new shock arrives.

Expectations are long-run expectations

Expectations in this dynamic model always refer to expectations about the value of variables in the long run. For example, the expected exchange rate E^e is the value of the actual exchange rate E that we expect will prevail in the long run. The short-run expected exchange rate E^e_{SR} is the expectation that we have while in the short run about the value E_{LR} that the exchange rate will take in the long run. In the long run, E_{LR} is constant over time and known, so $E^e_{LR} = E_{LR}$.

Short run and long run equilibria

There are two types of equilibria:

- In a **short-run equilibrium**
  1. The price level P remains fixed at its past value and does not change even when exogenous variables change;
  2. **Rational expectations**: $E^e_{SR} = E_{LR}$.
- In a **long-run equilibrium**
  1. Nominal variables have fully adjusted, so $E^e_{LR} = E_{LR}$ and the price level has reached its long-run value;
  2. Output Y is equal to the **full-employment level of output** Y^f.

The full-employment level of output is the same as potential output. Full-employment in this context does not mean that unemployment is zero. It means that the economy is operating at its long run "full capacity".

Shocks and how the economy evolves

We assume that, in the initial period, the economy is in a long-run equilibrium. All variables are constant over time and are expected to remain constant.

Then, we introduce a shock. As soon as the shock hits the economy, we are in the short run (by definition), and the economy is in a short-run equilibrium. No other shocks occur after the short run.

As time goes by, the economy's endogenous variables may keep adjusting in response to the shock. Once all variables stop changing and become constant, the economy reaches the long run and is in its final long-run equilibrium.

The period between the short run and the long run is called the "transition" or "adjustment" between the short run and the long run.

The adjustment between the short run and the long run can occur immediately, after an infinite amount of time, or anywhere in between. When the adjustment is immediate, the short run and the long run coincide, and the short run equilibrium is also a long run equilibrium.

Adjustment mechanism between the short run and the long run

Between the short run and the long run, the price level P adjusts according to the Phillips curve:

$$\pi_t = \alpha(Y_t - Y_t^f), \qquad \alpha > 0,$$

where $\pi_t = (P_t - P_{t-1})/P_{t-1}$ is the inflation rate.

When output is above the full-employment level ($Y_t > Y_t^f$), inflation is positive and the price level is increasing. A higher price level reduces real money balances M^s/P, shifting the AA curve down. At the same time, for a given nominal exchange rate, a higher P lowers the real exchange rate $q = EP^*/P$ (a real appreciation), CA deteriorates (goes down) and the DD curve shifts left. Both shifts reduce output, moving the economy back toward full employment.

When output is below the full-employment level ($Y_t < Y_t^f$), inflation is negative and the price level is decreasing. A lower price level shifts the AA curve up and the DD curve to the right. Both shifts raise output, moving the economy toward full employment.

When output equals the full-employment level ($Y_t = Y_t^f$), inflation is zero, the price level is constant, and neither the AA nor the DD curve shifts. The economy is in the long-run equilibrium.

Temporary and permanent shocks

A temporary increase in an exogenous variable X, shown in Figure 1.4, means

$$X_0 = X_{LR} = X_t \text{ for all } t > SR, \text{ and } X_{SR} > X_0.$$

A permanent increase in an exogenous variable X, shown in Figure 1.5, means

$$X_0 < X_{SR} = X_{LR} = X_t \text{ for all } t > SR.$$

Temporary and permanent decreases are defined analogously. The words "temporary" and "permanent" refer to the shock. The words "short run" and "long run" refer to the equilibrium concept.

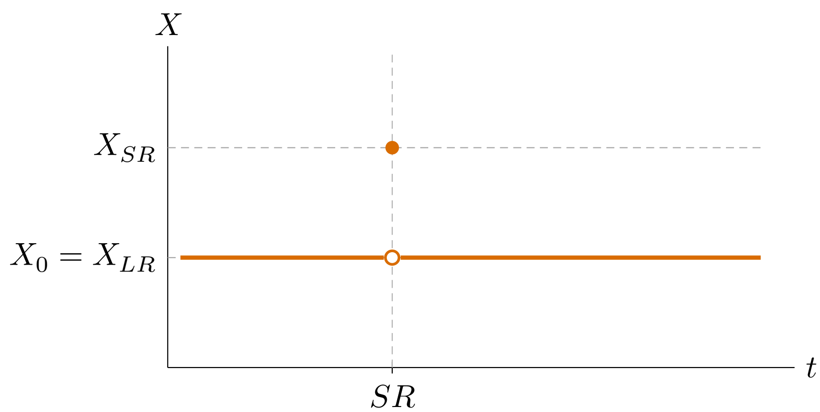

Figure 1.4: Time path of an exogenous variable X after a temporary increase. The variable jumps from X_0 to X_{SR} at the moment of the shock and immediately reverts to $X_0 = X_{LR}$.

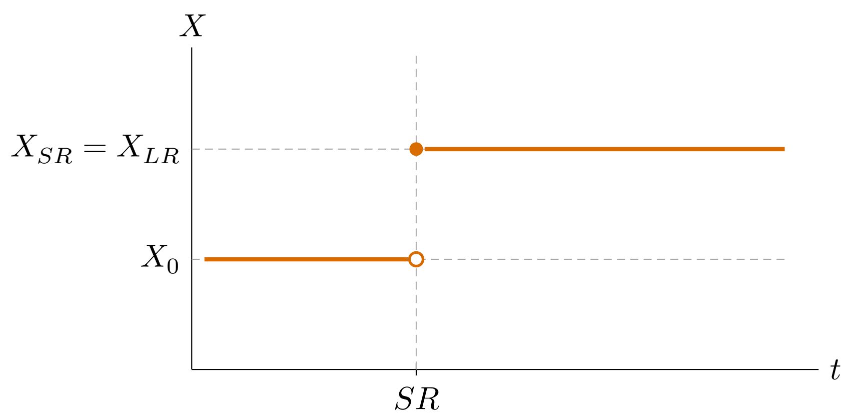

Figure 1.5: Time path of an exogenous variable X after a permanent increase. The variable jumps from X_0 to $X_{SR} = X_{LR}$ at the moment of the shock and remains at the new level.

1.8 How to solve the dynamic AA-DD model

The equations for the dynamic AA-DD model are:

AA curve:	$E_t = \text{AA}(Y_t; P_t, E_t^e, M_t^s, R_t^*), \text{ for all } t$
DD curve:	$E_t = \text{DD}(Y_t; P_t, T_t, I_t, G_t, Y_t^*, P_t^*), \text{ for all } t$
Initial LR equilibrium:	$E_0^e = E_0, \quad Y_0 = Y_0^f$
SR equilibrium:	$E_{SR}^e = E_{LR}, \quad P_{SR} = P_0$
Final LR equilibrium:	$E_{LR}^e = E_{LR}, \quad Y_{LR} = Y_{LR}^f$
Transition SR to LR:	$\pi_t = \alpha(Y_t - Y_t^f), \text{ for } t > SR$

We solve the model in this order:

1. Initial long-run equilibrium
2. Final long-run equilibrium
3. Short-run equilibrium
4. Transition between the short run and the long run

Initial long-run equilibrium

In the initial long-run equilibrium,

$$E_0^e = E_0 \quad \text{and} \quad Y_0 = Y_0^f. \tag{1.18}$$

Uncovered interest parity (equation (1.2)) with $E_0^e = E_0$ gives

$$R_0 = R_0^*. \tag{1.19}$$

Using (1.18) and (1.19) in the money market gives the initial price level:

$$P_0 = \frac{M_0^s}{\mathrm{L}(R_0^*, Y_0^f)}. \tag{1.20}$$

The goods market at full employment pins down the initial real exchange rate q_0:

$$Y_0^f = \mathsf{C}(Y_0^f - T_0) + I_0 + G_0 + \mathsf{CA}(q_0, Y_0^f - T_0, Y_0^*),$$

and once q_0 and P_0 are known, the initial nominal exchange rate follows from the definition of the real exchange rate:

$$E_0 = \frac{q_0 P_0}{P_0^*}.$$

Final long-run equilibrium

If the shock is temporary, all exogenous variables return to their initial values, so the final long-run equilibrium is the same as the initial long-run equilibrium.

If the shock is permanent, the final long-run equilibrium differs from the initial one. Following the same steps as for the initial long-run equilibrium, but with subscript LR instead of 0, we get $E_{LR}^e = E_{LR}$, $Y_{LR} = Y_{LR}^f$, $R_{LR} = R_{LR}^*$, and

$$\begin{aligned} P_{LR} &= \frac{M_{LR}^s}{\mathrm{L}(R_{LR}^*, Y_{LR}^f)}, \\ E_{LR} &= \frac{q_{LR} P_{LR}}{P_{LR}^*}. \end{aligned} \tag{1.21}$$

where q_{LR} solves

$$Y_{LR}^f = \mathsf{C}(Y_{LR}^f - T_{LR}) + I_{LR} + G_{LR} + \mathsf{CA}(q_{LR}, Y_{LR}^f - T_{LR}, Y_{LR}^*).$$

Short-run equilibrium

In a short-run equilibrium, prices are fixed and expectations are rational:

$$E^e_{SR} = E_{LR} \quad \text{and} \quad P_{SR} = P_0.$$

Since E_{LR} and P_0 are already known from the initial and final long-run equilibria, E^e_{SR} and P_{SR} are known. Substituting these into the AA and DD curves evaluated at $t = SR$ gives two equations in two unknowns, Y_{SR} and E_{SR}. Once Y_{SR} and E_{SR} are found, the interest rate R_{SR} can be found from UIP

$$R_{SR} = R^*_{SR} + \frac{E_{LR}}{E_{SR}} - 1.$$

The remaining short-run variables (R_{SR}, q_{SR}, C_{SR}, CA_{SR}) follow from their definitions.

Transition between the short run and the long run

If the shock is temporary, the exogenous variables revert to their initial values immediately after the short run. Since the final long-run equilibrium is the same as the initial long-run equilibrium, the economy returns to its initial state right away: there is no transition.

If the shock is permanent, the only adjustment mechanism between the short run and the long run is through the price level P. Whether adjustment occurs depends on whether $P_{LR} \neq P_0$.

Using equations (1.20) and (1.21):

$$P_{LR} = P_0 \quad \Longleftrightarrow \quad \frac{M^s_{LR}}{\mathrm{L}(R^*_{LR}, Y^f_{LR})} = \frac{M^s_0}{\mathrm{L}(R^*_0, Y^f_0)}.$$

Only a permanent shock to M^s, R^*, Y^f, or a parameter change in $\mathrm{L}(R, Y)$ can make $P_{LR} \neq P_0$. All other shocks (to T, I, G, Y^*, or P^*) leave $P_{LR} = P_0$.

If $P_{LR} = P_0$, the price level is constant throughout: $P_{SR} = P_0$ by sticky prices and $P_{LR} = P_0$ by the condition above. Since P is the only adjustment mechanism and it does not change, the AA and DD curves cannot shift after the short run. The economy reaches its final long-run equilibrium immediately, so the short-run and long-run equilibria coincide. Equivalently, with the price level constant, $\pi_t = 0$ and the Phillips curve implies $Y = Y^f$.

If $P_{LR} \neq P_0$, as described in Section 1.7, the Phillips curve drives P toward P_{LR}, shifting the AA and DD curves until $Y = Y^f_{LR}$ and inflation is zero. Throughout the transition, the expected exchange rate remains at E_{LR}, since expectations only change when there is a shock.

1.9 Policy shocks

We now apply the solution method to the four cases: a temporary monetary expansion, a temporary fiscal expansion, a permanent monetary expansion, and a permanent fiscal expansion. In each case, M^s or G changes while all other exogenous variables remain at their initial values. Since R^*, Y^f, and L are unchanged in all four cases, the condition $P_{LR} \neq P_0$ reduces to $M^s_{LR} \neq M^s_0$.

Temporary monetary expansion

The shock is $M^s_{SR} > M^s_0$ with $M^s_{LR} = M^s_0$.

Final long-run equilibrium. The shock is temporary, so the final long-run equilibrium is the same as the initial one. In particular, $E_{LR} = E_0$.

Short-run equilibrium. Since $E^e_{SR} = E_{LR} = E_0$, the AA curve shifts upward (higher M^s raises E for each Y) while the DD curve does not shift (see Figure 1.6). The economy moves from A to B: output rises above full employment and the domestic currency depreciates ($Y_{SR} > Y^f_0$ and $E_{SR} > E_0$).

Transition. The shock is temporary, so the economy returns to its initial long-run equilibrium immediately after the short run. There is no transition.

Temporary fiscal expansion

The shock is $G_{SR} > G_0$ with $G_{LR} = G_0$.

Final long-run equilibrium. The shock is temporary, so the final long-run equilibrium is the same as the initial one. In particular, $E_{LR} = E_0$.

Short-run equilibrium. Since $E^e_{SR} = E_{LR} = E_0$, the DD curve shifts to the right (higher G

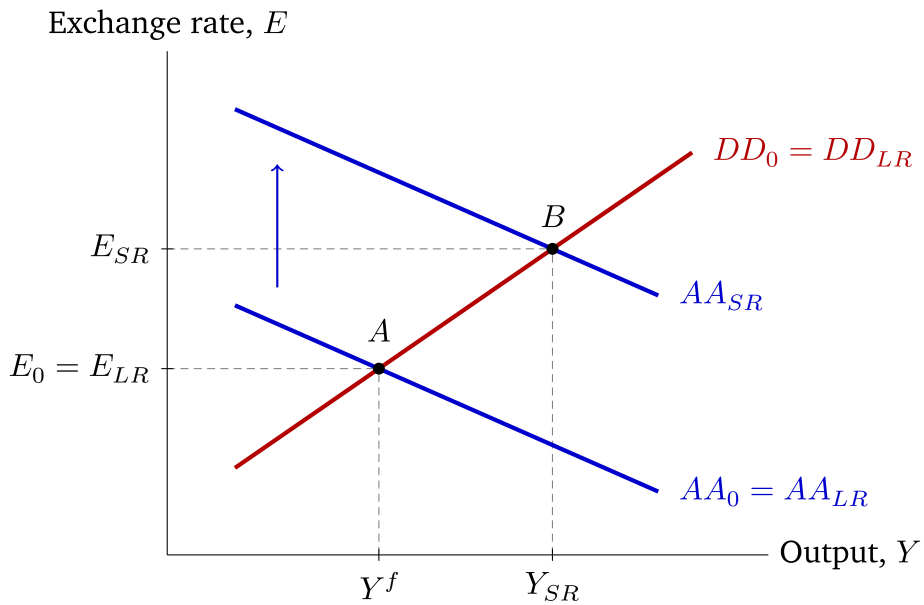

Figure 1.6: Temporary monetary expansion. The AA curve shifts upward from AA_0 to AA_{SR} while the DD curve does not shift. The economy moves from A to B: output rises and the domestic currency depreciates ($E_{SR} > E_0$). Because the shock is temporary, $AA_{LR} = AA_0$ and the economy eventually returns to A.

raises Y for each E) while the AA curve does not shift (see Figure 1.7). The economy moves from A to B: output rises above full employment and the domestic currency appreciates ($Y_{SR} > Y_0^f$ and $E_{SR} < E_0$).

Transition. The shock is temporary, so the economy returns to its initial long-run equilibrium immediately after the short run. There is no transition.

Permanent monetary expansion

The shock is $M_{SR}^s = M_{LR}^s > M_0^s$.

Final long-run equilibrium. Since $M_{LR}^s > M_0^s$ while R^* and Y^f are unchanged, the long-run price level rises: $P_{LR} > P_0$. The exogenous variables in the goods market are unchanged, so $q_{LR} = q_0$. With the real exchange rate unchanged, a higher long-run price level implies the long-run nominal exchange rate rises proportionally: $E_{LR}/E_0 = P_{LR}/P_0 = M_{LR}^s/M_0^s > 1$. In particular, $E_{LR} > E_0$.

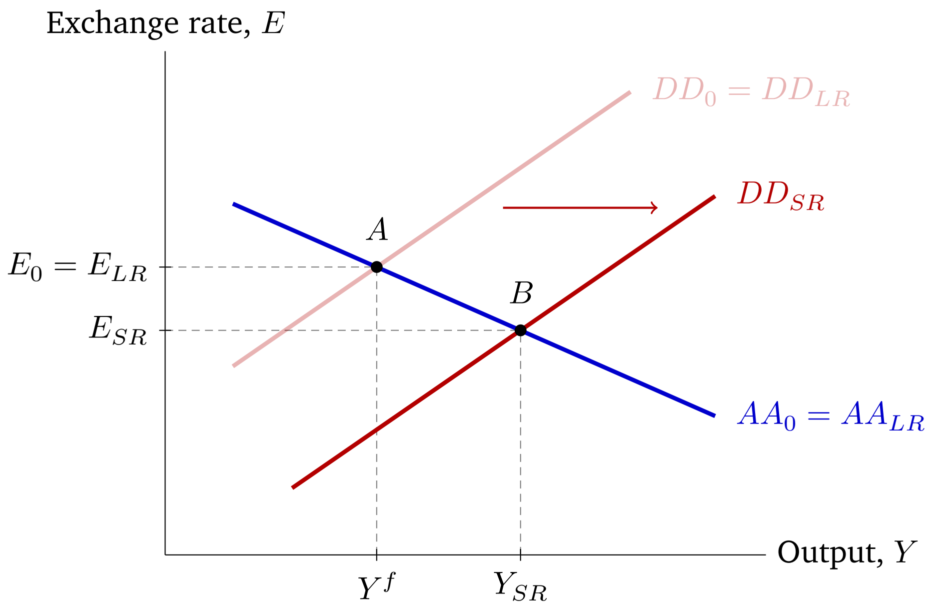

Figure 1.7: Temporary fiscal expansion. The DD curve shifts to the right from DD_0 to DD_{SR} while the AA curve does not shift. The economy moves from A to B: output rises and the domestic currency appreciates ($E_{SR} < E_0$). Because the shock is temporary, $DD_{LR} = DD_0$ and the economy eventually returns to A.

Short-run equilibrium. Since $E^e_{SR} = E_{LR} > E_0$, the AA curve shifts upward by more than in the temporary case (the expected depreciation reinforces the effect of higher M^s). The DD curve does not shift in the SR since $P_{SR} = P_0$ and none of the goods market exogenous variables changed. The economy moves from A to B in Figure 1.8: output rises above full employment and the domestic currency depreciates ($Y_{SR} > Y^f_0$ and $E_{SR} > E_0$). In fact, $E_{SR} > E_{LR}$: the exchange rate *overshoots* its long-run level.

Transition. Since $P_{LR} > P_0$ and $Y_{SR} > Y^f_0$, inflation is positive and P rises gradually from P_0 toward P_{LR}. The AA curve shifts downward and the DD curve shifts leftward until $Y = Y^f_{LR}$ and $E = E_{LR}$. Figure 1.9 shows the transition path from B to C.

Permanent fiscal expansion

The shock is $G_{SR} = G_{LR} > G_0$.

Final long-run equilibrium. Since M^s, R^*, and Y^f are unchanged, $P_{LR} = P_0$. The higher G raises demand for domestic goods, so equilibrium in the goods market at full employment

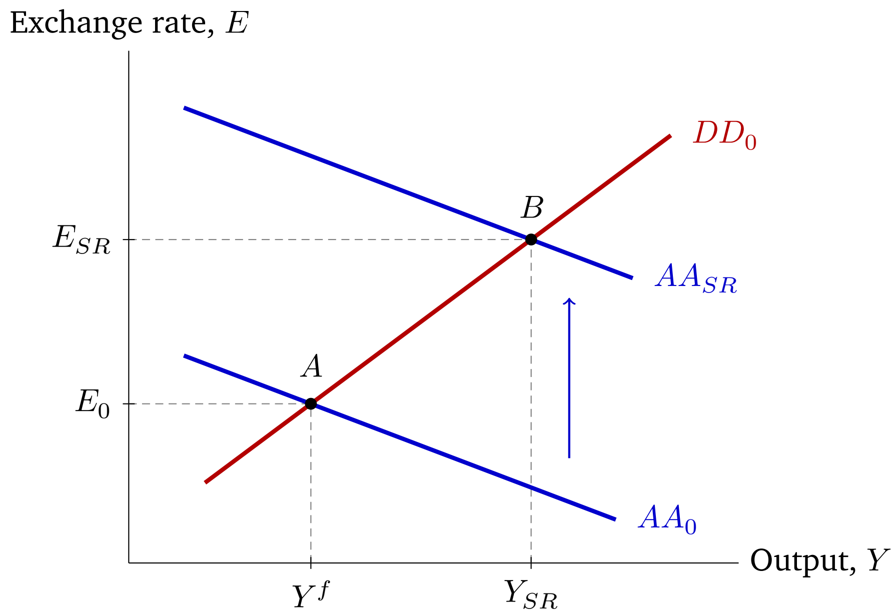

Figure 1.8: Permanent monetary expansion: initial impact. The AA curve shifts upward from AA_0 to AA_{SR}; the DD curve does not shift on impact. The economy jumps from A to B. The exchange rate overshoots: $E_{SR} > E_{LR} > E_0$.

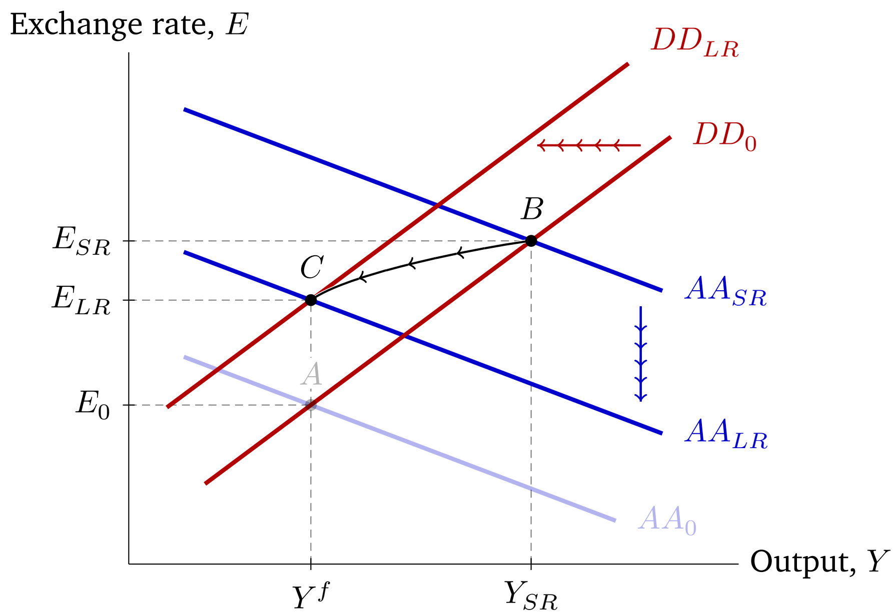

Figure 1.9: Permanent monetary expansion: transition from short run to long run. Starting from B, positive inflation shifts the AA curve downward (from AA_{SR} toward AA_{LR}) and the DD curve leftward (from DD_0 toward DD_{LR}). The economy follows the curved path from B to C, where C is the long-run equilibrium at (Y^f, E_{LR}). Point A (faded) marks the original equilibrium.

requires a lower q_{LR}: the real exchange rate appreciates ($q_{LR} < q_0$). Since $P_{LR} = P_0$ and P^* is unchanged, the nominal exchange rate appreciates as well: $E_{LR} < E_0$.

Short-run equilibrium. Since $E^e_{SR} = E_{LR} < E_0$, the AA curve shifts downward (the expected appreciation lowers the dollar return on foreign bonds, requiring a lower E for each Y). The DD curve shifts to the right (higher G). As shown in Figure 1.10, the economy moves from A to B. The appreciation lowers the current account one for one relative to the increase in G.

Transition. Since $P_{LR} = P_0$, there is no transition. The short-run equilibrium coincides with the final long-run equilibrium: $Y_{SR} = Y^f_0$, $E_{SR} = E_{LR}$.

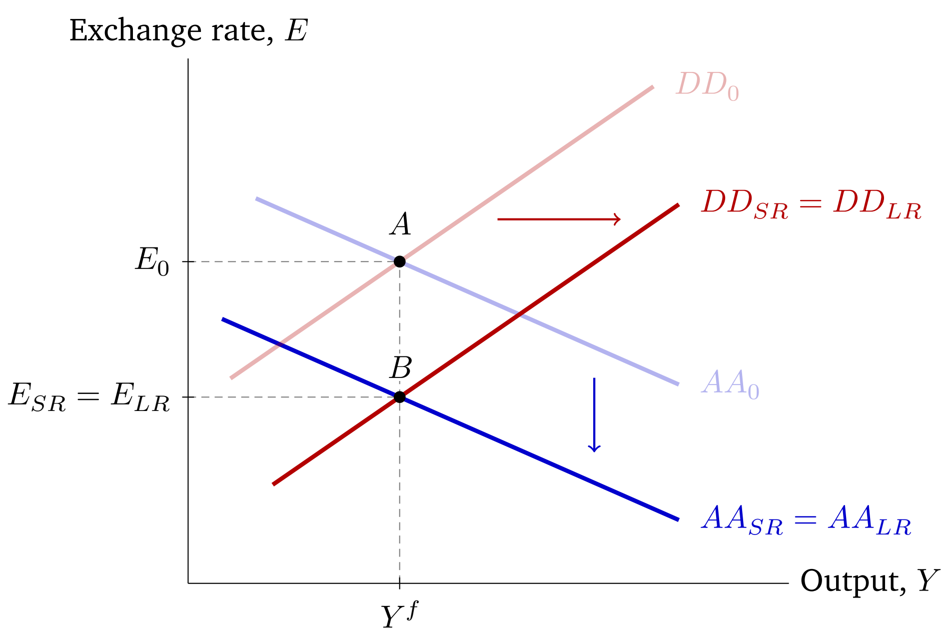

Figure 1.10: Permanent fiscal expansion. Both curves shift: the DD curve shifts to the right and the AA curve shifts downward. The economy moves from A to B, where B is both the short-run and long-run equilibrium. Output is unchanged at Y^f; the domestic currency appreciates permanently ($E_{SR} = E_{LR} < E_0$).

1.10 A numerical example

We now work through a numerical example that illustrates the four cases. We use the same functional forms introduced in the earlier examples: consumption from Example 1.6, the current account from Example 1.5, and money demand from Example 1.8:

$$\begin{aligned} \mathrm{C}(Y^d) &= \frac{1}{2}Y^d, \\ \mathrm{CA}(q, Y^d, Y^*) &= -\frac{1}{10} + \frac{1}{5}q - \frac{1}{10}Y^d + \frac{1}{10}Y^*, \\ \mathrm{L}(R, Y) &= \frac{Y}{1+R}. \end{aligned}$$

The demand for domestic goods D was derived in Example 1.6, the DD curve in Example 1.9, and the AA curve in Example 1.8.

Common specification and initial equilibrium

The initial exogenous values are

$$\begin{aligned} &T_0 = 0, && I_0 = \frac{1}{5}, && G_0 = \frac{1}{5}, && R_0^* = \frac{2}{25}, \\ &P_0^* = \frac{27}{25}, && Y_0^f = 1, && Y_0^* = 1, && M_0^s = 1. \end{aligned}$$

Using these values, the demand function from Example 1.6 becomes

$$\mathrm{D}(q, Y, 0, 1/5, 1/5) = \frac{2}{5} + \frac{1}{5}q + \frac{2}{5}Y.$$

Following the steps for the initial long-run equilibrium: $Y_0 = Y_0^f = 1$ gives $q_0 = 1$ from the goods market; $R_0 = R_0^* = 2/25$ from UIP; $P_0 = 27/25$ from the money market; and $E_0 = q_0 P_0 / P_0^* = 1$. Therefore,

$$(Y_0, q_0, P_0, E_0, R_0) = \left(1, 1, \frac{27}{25}, 1, \frac{2}{25}\right).$$

The implied initial current account is $CA_0 = \mathrm{CA}(1, 1, 1) = 1/10 > 0$.

From the DD curve in Example 1.9 and the money market in Example 1.8, two DD curves and two short-run money-market relations will recur below:

$$\begin{aligned} E &= -2 + 3Y && \text{when } G_{SR} = \frac{1}{5}, \\ E &= -\frac{9}{4} + 3Y && \text{when } G_{SR} = \frac{1}{4}, \\ R &= -1 + \frac{27}{25}Y && \text{when } M^s_{SR} = 1, \\ R &= -1 + \frac{36}{35}Y && \text{when } M^s_{SR} = \frac{21}{20}. \end{aligned}$$

Temporary monetary expansion

The shock is $M^s_{SR} = 21/20$ with $M^s_{LR} = M^s_0 = 1$ and $G_{SR} = G_{LR} = G_0 = 1/5$.

Final long-run equilibrium. The shock is temporary, so the final long-run equilibrium equals the initial one:

$$(q_{LR}, P_{LR}, E_{LR}, R_{LR}) = \left(1, \frac{27}{25}, 1, \frac{2}{25}\right).$$

Short-run equilibrium. Since $E^e_{SR} = E_{LR} = 1$ and $P_{SR} = P_0 = 27/25$, the DD curve, money market, and UIP give

$$E = -2 + 3Y, \qquad R = -1 + \frac{36}{35}Y, \qquad R = -\frac{23}{25} + \frac{1}{E}.$$

Solving yields $Y_{SR} = 1.01$, $E_{SR} = 1.04$, $R_{SR} = 0.04$.

Transition. The shock is temporary, so the economy returns to its initial equilibrium immediately. There is no transition.

The remaining endogenous variables appear in Tables 1.1 and 1.2.

Temporary fiscal expansion

The shock is $G_{SR} = 1/4$ with $G_{LR} = G_0 = 1/5$ and $M^s_{SR} = M^s_{LR} = M^s_0 = 1$.

Final long-run equilibrium. The shock is temporary, so the final long-run equilibrium equals the initial one:

$$(q_{LR}, P_{LR}, E_{LR}, R_{LR}) = \left(1, \frac{27}{25}, 1, \frac{2}{25}\right).$$

Short-run equilibrium. Since $E^e_{SR} = E_{LR} = 1$ and $P_{SR} = P_0 = 27/25$, the DD curve, money market, and UIP give

$$E = -\frac{9}{4} + 3Y, \qquad R = -1 + \frac{27}{25}Y, \qquad R = -\frac{23}{25} + \frac{1}{E}.$$

Solving yields $Y_{SR} = (89 + \sqrt{19729})/216 = 1.06$, $E_{SR} = 0.94$, $R_{SR} = 0.15$.

Transition. The shock is temporary, so the economy returns to its initial equilibrium immediately. There is no transition.

The remaining endogenous variables appear in Tables 1.1 and 1.2.

Permanent monetary expansion

The shock is $M^s_{SR} = M^s_{LR} = 21/20$ with $G_{SR} = G_{LR} = G_0 = 1/5$.

Final long-run equilibrium. The goods block is unchanged, so $q_{LR} = q_0 = 1$. The money market gives $P_{LR} = M^s_{LR}/\mathrm{L}(R^*_{LR}, Y^f_{LR}) = 567/500 > P_0$. Then $R_{LR} = R^*_{LR} = 2/25$ and $E_{LR} = q_{LR}P_{LR}/P^*_{LR} = 21/20 > E_0$:

$$(q_{LR}, P_{LR}, E_{LR}, R_{LR}) = \left(1, \frac{567}{500}, \frac{21}{20}, \frac{2}{25}\right).$$

Short-run equilibrium. Since $E^e_{SR} = E_{LR} = 21/20$ and $P_{SR} = P_0 = 27/25$, the DD curve, money market, and UIP give

$$E = -2 + 3Y, \qquad R = -1 + \frac{36}{35}Y, \qquad R = -\frac{23}{25} + \frac{21}{20E}.$$

Solving yields $Y_{SR} = (67 + \sqrt{13834})/180 = 1.03$, $E_{SR} = 1.08$, $R_{SR} = 0.06$. Since $E_{SR} = 1.08 > E_{LR} = 1.05$, the exchange rate overshoots.

Transition. Since $P_{LR} = 567/500 > P_0 = 27/25$ and $Y_{SR} > Y^f_0$, inflation is positive and P rises from P_0 toward P_{LR}, shifting the AA curve downward and the DD curve leftward until $Y = Y^f_{LR} = 1$ and $E = E_{LR} = 21/20$.

The remaining endogenous variables appear in Tables 1.1 and 1.2.

Permanent fiscal expansion

The shock is $G_{SR} = G_{LR} = 1/4$ with $M^s_{SR} = M^s_{LR} = M^s_0 = 1$.

Final long-run equilibrium. Since M^s, R^*, and Y^f are unchanged, $P_{LR} = P_0 = 27/25$. The goods market at full employment gives $q_{LR} = 3/4 < q_0$. Then $E_{LR} = q_{LR}P_{LR}/P^*_{LR} = 3/4 < E_0$:

$$(q_{LR}, P_{LR}, E_{LR}, R_{LR}) = \left(\frac{3}{4}, \frac{27}{25}, \frac{3}{4}, \frac{2}{25}\right).$$

Short-run equilibrium. Since $E^e_{SR} = E_{LR} = 3/4$ and $P_{SR} = P_0 = 27/25$, the DD curve, money market, and UIP give

$$E = -\frac{9}{4} + 3Y, \qquad R = -1 + \frac{27}{25}Y, \qquad R = -\frac{23}{25} + \frac{3}{4E}.$$

Solving gives $Y_{SR} = 1$, $E_{SR} = 3/4$, $R_{SR} = 2/25$. The short-run equilibrium coincides with the final long-run equilibrium.

Transition. Since $P_{LR} = P_0$, there is no transition.

The remaining endogenous variables appear in Tables 1.1 and 1.2.

Tables 1.1 and 1.2 summarize the numerical results. All decimal entries are rounded to two digits after the decimal point.

state	Y	Y^d	C	CA	D
Temporary monetary expansion					
0	1.00	1.00	0.50	0.10	1.00
SR	1.01	1.01	0.51	0.11	1.01
LR	1.00	1.00	0.50	0.10	1.00
Temporary fiscal expansion					
0	1.00	1.00	0.50	0.10	1.00
SR	1.06	1.06	0.53	0.08	1.06
LR	1.00	1.00	0.50	0.10	1.00
Permanent monetary expansion					
0	1.00	1.00	0.50	0.10	1.00
SR	1.03	1.03	0.51	0.11	1.03
LR	1.00	1.00	0.50	0.10	1.00
Permanent fiscal expansion					
0	1.00	1.00	0.50	0.10	1.00
SR	1.00	1.00	0.50	0.05	1.00
LR	1.00	1.00	0.50	0.05	1.00

Table 1.1: Real-side outcomes in the numerical example.

state	q	P	E^e	E	R	M^d
Temporary monetary expansion						
0	1.00	1.08	1.00	1.00	0.08	1.00
SR	1.04	1.08	1.00	1.04	0.04	1.05
LR	1.00	1.08	1.00	1.00	0.08	1.00
Temporary fiscal expansion						
0	1.00	1.08	1.00	1.00	0.08	1.00
SR	0.94	1.08	1.00	0.94	0.15	1.00
LR	1.00	1.08	1.00	1.00	0.08	1.00
Permanent monetary expansion						
0	1.00	1.08	1.00	1.00	0.08	1.00
SR	1.08	1.08	1.05	1.08	0.06	1.05
LR	1.00	1.13	1.05	1.05	0.08	1.05
Permanent fiscal expansion						
0	1.00	1.08	1.00	1.00	0.08	1.00
SR	0.75	1.08	0.75	0.75	0.08	1.00
LR	0.75	1.08	0.75	0.75	0.08	1.00

Table 1.2: Price, exchange-rate, and money-market outcomes in the numerical example. The final column reports nominal money demand, $M^d = P\mathbb{L}(R, Y)$.

1.11 Summary of AA and DD shifts

Table 1.3 summarizes how an increase in each exogenous variable shifts the AA and DD curves. A decrease in the variable shifts the relevant curve in the opposite direction.

Increase in	AA	DD
P	down	left
E^e	up	no shift
M^s	up	no shift
R^*	up	no shift
T	no shift	left
I	no shift	right
G	no shift	right
Y^*	no shift	right
P^*	no shift	right

Table 1.3: Direction of AA and DD shifts when an exogenous variable increases.

Table 1.4 summarizes how a *permanent* increase in an exogenous variable affects the long-run price level. Temporary shocks do not change the final long-run price level because all exogenous variables return to their initial values. Permanent changes in T, I, G, Y^*, and P^* do not change P_{LR}.

Permanent increase in	P_{LR}
M^s	up
R^*	up
Y^f	down
parameter change in L(R, Y)	depends on change
T, I, G, Y^*, P^*	no change

Table 1.4: Direction of the long-run price-level response to a permanent increase in an exogenous variable.